Solutions and teaching notes, chapter 10

Teacher use. The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. 2π. *Action:* Link radius, diameter and perimeter.
2. $90°$. *Action:* Divide a circle into four sectors.
3. $\dfrac{\sqrt{2}}{2}$. *Action:* Rationalise and check the square.
4. $(a;-b)$. *Action:* Plot both points.
5. $-\dfrac{\pi}{3}$. *Action:* Put over a common denominator.
6. $\dfrac{3}{5}$. *Action:* Identify the reference angle.

Investigation

Equilateral triangle then right-angled isosceles triangle. Pythagoras gives the height $\dfrac{\sqrt{3}}{2}$ and equal coordinates $\dfrac{\sqrt{2}}{2}$. The signs are positive in the first quadrant.

Direct applications

1. $\dfrac{\pi}{4}$, $\dfrac{2\pi}{3}$; length 3 $\dfrac{\pi}{2}$ m.
2. $\dfrac{7\pi}{2}$ has the image of $3\dfrac{\pi}{2}$; $-\dfrac{5\pi}{3}$ that of $\dfrac{\pi}{3}$.
3. $\sin^2 x = \dfrac{16}{25}$; the sine is negative, so $-\dfrac{4}{5}$.
4. $(-\dfrac{1}{2};-\dfrac{\sqrt{3}}{2})$ and $(\dfrac{\sqrt{2}}{2};-\dfrac{\sqrt{2}}{2})$.
5. $h = \dfrac{\sqrt{3}}{2}$; new chord $\dfrac{1}{\sqrt{2+\sqrt{3}}} = \sqrt{2-\sqrt{3}}$.

Exercises: methods and answers

Exercise 10.1 · Convert and measure. $\dfrac{5\pi}{6}$ and $-\dfrac{\pi}{2}$; arc $\dfrac{4\pi}{3}$ cm. The negative sign indicates a orientation; a geometric length is positive.

Exercise 10.2 · Reduce angles. $\dfrac{5\pi}{6}$, $\dfrac{\pi}{4}$, π. Remove or add multiples of 2 π.

Exercise 10.3 · Exact values. $(-\frac{1}{2};\frac{\sqrt{3}}{2})$; $(\frac{\sqrt{3}}{2};-\frac{1}{2})$; $(-\frac{\sqrt{2}}{2};-\frac{\sqrt{2}}{2})$, in the form (cosine ; sine).

Exercise 10.4 · Find the sign. $\cos^2 x = 1-\frac{25}{169}=\frac{144}{169}$; negative cosine, so $-\frac{12}{13}$.

Exercise 10.5 · A wheel. Distance travelled $r\theta = 0{,}3(\frac{5\pi}{3})=\frac{\pi}{2}$ m. This corresponds to unrolling the arc; the chord joins two positions on the circle and does not have this length. A point fixed on the wheel traces a different path relative to the ground.

Exercise 10.6 · Chord and scalar product. $A(1;0)$, $B(-\frac{1}{2};\frac{\sqrt{3}}{2})$; $AB^2=\frac{9}{4}+\frac{3}{4}$ = 3. By Al-Kashi : $1+1-2\cos(\frac{2\pi}{3})$ = 3. Hence $AB=\sqrt{3}$.

Exercise 10.7 · Read an equation on the circle. The vertical line with x-coordinate $\frac{1}{2}$ intersects the circle at the angles $\frac{\pi}{3}$ and $\frac{5\pi}{3}$. The two points have opposite sines.

Exercise 10.8 · Delayed review. $\frac{25\pi}{4}=6\pi+\frac{\pi}{4}$. Three full turns do not change the image, so both values are equal to $\frac{\sqrt{2}}{2}$.

CQFD, exercise 32 p. 220, questions a and b. a : $\cos(\frac{6\pi}{5})=\cos(\pi+\frac{\pi}{5})=-\frac{1+\sqrt{5}}{4}$. b : $\cos(-\frac{\pi}{5})=\cos(\frac{\pi}{5})=\frac{1+\sqrt{5}}{4}$, since cosine is even.

CQFD, Exercise 39 p. 220, Questions 1, 2, 3 and 5. 1 : $\frac{\pi}{6}+\frac{\pi}{3}=\frac{\pi}{2}$. 2 : yes, $\cos(\frac{\pi}{3})=\frac{1}{2}=0{,}5$ is a decimal. 3: the principal measure in] $-\pi;\pi$] is $-\frac{5\pi}{6}$. 5 : $\sin(\frac{4\pi}{3})=-\frac{\sqrt{3}}{2}$.

MCQ: answers and reasons

MCQ 1.

1. B. Half-turn.

2. A. Equilateral triangle.

3. D. Lowest point of the circle.

4. C. Equation of the unit circle.

5. B. Complete turns.

MCQ 2.

1. C. Vertical symmetry.

2. A. Isosceles right triangle.

3. D. Three-quarters of a turn.

4. B. $r\theta = 6$.

5. C. Negative x-coordinate inside the quadrant.

True or false

1. False: at 0, the two sides would be 1 and -1. Cosine is even.

2. True: a complete turn keeps the image.

3. False: $\frac{\pi}{6}$ and $5\frac{\pi}{6}$ have the same sine but opposite cosines.

Differentiated teaching. If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.